Project Proposal and Methodology

Assignment: Hybrid RAG & Fine-Tuning for Customer Support

Stage 1: Business Objective and Proposed Methodology

Task 1.1: Define Project Scope

1.1.1 Identify the Target Use Case

This project focuses on policy-grounded billing dispute resolution for e-commerce customer support. It will process a free-form payment-related query, identify the underlying intent, retrieve the most relevant Standard Operating Procedure (SOP), and generate a response grounded in that policy.

Billing disputes were selected because customers often describe duplicate charges, incorrect amounts, failed payments, unrecognised transactions, or payments deducted without order confirmation using incomplete or emotional language. A general-purpose model may answer fluently while inventing reversal timelines, refund guarantees, investigation steps, or escalation routes. The proposed system therefore combines structured intent extraction with retrieval.

The business-facing use case is Billing Disputes, while PAYMENT will be used as the standardised dataset category. Within the PAYMENT category, the selected dataset intents are payment_issue and check_payment_methods, with payment_issue serving as the primary billing-dispute intent. The primary policy document is billing_disputes.md; payment_methods.md, refund_policy.md, and escalation_matrix.md may provide supporting context where confirmed during SOP inspection. The complete SOP collection will remain in the retrieval corpus so the system must select the relevant policy from both related and unrelated documents.

Three systems will be compared: a pretrained baseline model without retrieval, a naive RAG system using the raw query for retrieval, and a Hybrid RAG system in which a fine-tuned router produces structured JSON before retrieval.

Expected Input and Outputs

```
Input: {"query": "My card was charged twice for the same order. What should I do?"}
Router output: {"intent": "payment_issue", "category": "PAYMENT"}
Final output: {"intent": "payment_issue", "category": "PAYMENT", "response": "<SOP-grounded guidance>",
"policy_source": "billing_disputes.md"}
```

The router must return valid JSON only, with lowercase intent values and an uppercase category. The final response must be supported by retrieved SOP content and must not invent payment timelines, refund promises, contact details, investigation requirements, or escalation procedures.

Project Boundaries

- English-language, single-turn queries focused on payment issues and payment-method enquiries.
- Policy guidance only; no access to customer accounts, bank records, payment gateways, or order systems.
- No approval, rejection, reversal, refund, or investigation of real transactions.
- The supplied SOP corpus is the authoritative source; unrelated issues are outside scope or routed for escalation.
- LoRA or QLoRA will be used, with the fine-tuned model acting primarily as an intent router.
- Multi-turn dialogue, multilingual support, and live system integration are outside scope.

1.1.2 Define Success Criteria

The final Hybrid RAG system should show measurable improvement over both the baseline and naive RAG systems. All systems will be evaluated using the same held-out test set, SOP corpus, inference settings, and references.

Metric	Target
JSON format adherence	≥ 95%
Intent accuracy	≥ 85%
Adversarial intent accuracy	≥ 75%
Output consistency	≥ 95%
ROUGE-1	≥ 0.40 or ≥ 10% over baseline
ROUGE-L	≥ 0.30 or ≥ 10% over baseline
BLEU	≥ 0.15 or ≥ 10% over baseline
Hallucination frequency	< 10%
Hallucination reduction	≥ 40% relative to baseline

Format adherence will test whether router outputs are valid and schema-compliant JSON. Intent accuracy will use exact matching against held-out labels. ROUGE and BLEU will compare generated answers with SOP-grounded references, and consistency will be measured through repeated deterministic inference. A response will be marked hallucinated when it contains any unsupported payment timeline, refund promise, investigation step, escalation route, contact detail, or other claim absent from the retrieved SOP context.

Task 1.4: Design Methodology

1.4.1 Select Baseline Model

The selected baseline model is Qwen/Qwen2.5-1.5B-Instruct. It is an open-source, instruction-tuned model that offers a practical balance between generation quality, structured-output capability, and Google Colab GPU constraints. Its 1.5-billion-parameter size is suitable for 4-bit inference and parameter-efficient fine-tuning, whereas a substantially larger model would increase memory usage, training time, and the risk of runtime failure.

The same base model and tokenizer will be retained for the baseline, naive RAG, and Hybrid RAG systems. This controls the experiment so that performance changes can be attributed to retrieval and fine-tuning rather than to a different language model. Billing disputes within the PAYMENT category will remain the principal business use case and the focus of detailed error analysis.

Baseline inference will be deterministic, using do_sample=False and temperature=None. This avoids sampling variation and supports fair consistency comparisons across all three system versions.

1.4.2 Design Retrieval Strategy

The retrieval knowledge base will contain the complete set of corporate SOP Markdown documents. Each document will be cleaned, divided into retrieval-ready chunks, and stored with metadata such as source filename and chunk identifier.

Semantic embeddings will be generated using sentence-transformers/all-MiniLM-L6-v2. This model is computationally efficient and appropriate for a relatively small policy corpus. The embeddings will be stored in ChromaDB and retrieved using cosine similarity. The initial configuration will return the top three most similar chunks for each query.

Retrieval Component	Planned Configuration
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector store	ChromaDB
Similarity measure	Cosine similarity
Initial retrieval depth	Top-k = 3

In Solution V1, the raw customer query will be used directly for semantic retrieval. The retrieved chunks will then be inserted into the generation prompt as supporting context. In Solution V2, the fine-tuned router will first extract the category and intent. These values will be combined with the original query to form a more explicit search string, for example: “Category: PAYMENT; Intent: payment_issue; Customer issue: card charged twice.”

Retrieval quality will be assessed by checking whether the expected SOP appears within the top-k results, reviewing similarity scores, and analysing failed retrieval cases. The embedding model, chunking method, top-k value, and vector-store configuration will remain fixed between Solution V1 and Solution V2.

1.4.3 Design Fine-Tuning Strategy

The model will be fine-tuned as an intent router using QLoRA. QLoRA combines 4-bit quantisation with trainable low-rank adapters, reducing GPU-memory requirements while leaving the original model weights frozen. This is suitable for Google Colab and for the structured JSON intent-extraction task.

Parameter	Initial Configuration
Quantisation	4-bit NF4
LoRA rank	16
LoRA alpha	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj
Optimiser	AdamW
Initial learning rate	2×10^{-4}
Training duration	Maximum 3 epochs

The exact batch size, gradient-accumulation steps, and maximum sequence length will be finalised after token-length analysis in the Data Preparation notebook. Each record will be converted to ChatML. The user message will contain the raw customer query, and the assistant message will contain only the target JSON, for example: {"intent": "payment_issue", "category": "PAYMENT"}.

Label masking with -100 will be applied to system and user tokens so that training loss is calculated only on the assistant JSON response. Training and validation loss will be recorded at regular intervals. The checkpoint with the lowest validation loss will be retained, early stopping will be applied to limit overfitting, and train-versus-validation loss curves and training logs will be preserved as reproducibility evidence.

1.4.4 Define Evaluation Framework

The baseline, naive RAG, and Hybrid RAG systems will be evaluated using the same held-out test split, SOP corpus, model configuration, and deterministic inference settings. Training, validation, and test records will remain independent.

- Format Adherence: parse router outputs as JSON and verify the required intent and category fields, with no explanatory text outside the JSON object.
- Intent Accuracy: compare the predicted intent with the held-out ground-truth intent using exact-match accuracy.
- Response Quality: calculate ROUGE-1, ROUGE-L, and BLEU against SOP-grounded reference responses.
- Consistency: repeat deterministic inference for the same query and measure whether the intent and policy guidance remain stable.
- Hallucination Frequency: identify unsupported timelines, refund promises, payment procedures, escalation routes, contact details, or other claims absent from the retrieved SOP context.
- Adversarial Evaluation: evaluate a held-out subset containing emotional, indirect, negative, hedged, or noisy customer language.
- Retrieval Quality: verify whether the expected source document appears within the top-k retrieved results.

The final comparative analysis will report absolute scores and relative improvements across the baseline, Solution V1, and Solution V2. This will show the independent contribution of retrieval and fine-tuning to intent routing, grounding, consistency, and hallucination reduction.